

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I V.C. BHAVAN BAIRAV ¹⁹²⁵¹¹³⁶⁹ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.C. Bhuvan Bairav

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1) A company experiences unpredictable traffic spikes during seasonal sales.

1) Auto Scaling and Load Balancing:

- Auto Scaling automatically adds or removes servers based on traffic.
- Load Balancing distributes user requests evenly across multiple servers.
- Prevents server overload during seasonal sales.

2) Reducing Downtime and Improving User Experience:

- Automatically increases resources during traffic spikes.
- Ensures the website remains available without crashes.
- Reduces response time and page loading delays.
- Provides a smooth shopping experience with minimal downtime.

3) Role of Monitoring Tools:

- Continuously monitor CPU, memory and network usage.
- Detect high traffic and trigger auto scaling.
- Send alerts for server failures or performance issues.
- Help administrators identify and resolve problems quickly.

Eg. During Amazon or Flipkart festive sales, millions of users access the website simultaneously.

- Auto-Scaling launches additional servers, while load balancing distributes traffic evenly.

2) An organization wants to store large volumes of backup data securely in the cloud.

1) Cloud storage, Durability and Data Redundancy:

- Store backup data in cloud storage.
- Maintain multiple copies of data in different locations.
- Use data redundancy to prevent data loss.
- Ensure high durability and availability.

2) Encryption and Access Control:

- Encrypt backup data during storage and transmission.
- Use Role Based Access Control (RBAC) to restrict unauthorized access.
- Enable Multi-Factor Authentication (MFA) for better security.
- Allow access only to authorized users.

3) Cost Advantages:

- No need to purchase and maintain physical storage devices.
- Use pay-as-you-go pricing.
- Easily Scale storage based on requirements.
- Lower maintenance and operational costs.

4) Disaster Recovery Example:

- If a company's data center fails due to a fire or cyberattack, backup data stored in the cloud can be restored quickly.
- This ensures minimal data loss, quick recovery and business continuity.

3) A mobile app company plans to deploy its application globally.

1) CDN and Performance:

- Use a Content Delivery Network (CDN) to deliver content from the nearest server.
- Reduce Latency by serving data closer to users.
- Improve application speed and performance.

2) Role of Edge Servers:

- Edge servers store cached near users.
- Reduce the distance data travels.
- Decrease Response time and improve user experience.
- Minimise Network congestion.

3) Global Availability:

- Cloud Providers deploy applications across multiple regions.
- Use load balancing and auto-scaling to handle user traffic.
- Ensure continuous service even during high demand or server failures.

4) Real-World Example:

- Netflix uses CDNs and edge servers to stream videos worldwide.
- Users receive content from the nearest server, resulting in low latency, faster streaming and minimal buffering.

4) A financial institution requires strict compliance and data security in cloud usage.

1) Regulatory Compliance and Auditing:

- cloud providers comply with security standards and regulations.
- Maintain audit logs to track all user activities.
- Perform regular security audits and compliance checks.
- Ensure financial data is handled securely.

2) Logging, Monitoring and Encryption:

- Logging records all system and user activities.
- Monitoring detects suspicious activities and security threats.
- Encryption protects data both at rest and during transmission.
- Enable Multi-Factor Authentication (MFA) for secure access.

3) Data Sovereignty:

- Store Financial data in approved geographic regions.
- Choose the cloud regions that comply with government regulation.
- Restrict unauthorized access using Role-Based Access Control (RBAC)

4) Example:

- In an online banking system, customer transaction data is encrypted, user activities are logged and continuous monitoring detects fraud.
- This ensures regulatory compliance, data security, and protection of sensitive financial information.

5) A development team wants to adopt containerization for application deployment.

1) Containers V/s Virtual Machines:

- Containers share the host operating system, making them lightweight and fast.
- Virtual Machines (VMs) have their own operating system, consuming more resources.
- Containers start quickly and use less CPU and memory than VMs.

2) Role of Kubernetes:

- Automates container deployment and management.
- Performs load balancing and auto scaling.
- Restarts failed containers automatically.
- Simplifies application updates and maintenance.

3) Portability across clouds:

- Containers run consistently on AWS, Azure and Google cloud.
- Applications can be moved between cloud providers without major changes.
- Reduces vendor lock-in and improves flexibility.

Example:

An e-commerce application with separate services for login, payment, product catalog, and orders uses Kubernetes to deploy and scale containers efficiently across different cloud environments.